

Common GIT Diseases

Ulcerative Dyspepsia

Ulcerative colitis (UC) is an inflammatory bowel disease. It causes irritation, inflammation, and ulcers in the lining of large intestine. The overall estimated IBD population in India in 2010 came out to be 1.4 million, which was the second highest number after the USA (with 1.64 million). Therefore, although the disease prevalence in India is lower than in the West, with a population of more than 120 million, the total IBD population in India is among the largest across the globe.

The primary symptoms of active disease are abdominal pain and diarrhea mixed with blood. Weight loss, fever, and anemia may also occur. Often, symptoms come on slowly and can range from mild to severe. Symptoms typically occur intermittently with periods of no symptoms between flares. Complications may include megacolon, inflammation of the eye, joints, or liver, and colon cancer.

The cause of UC is unknown. Theories involve immune system dysfunction, genetics, changes in the normal gut bacteria, and environmental factors. Rates tend to be higher in the developed world with some proposing this to be the result of less exposure to intestinal infections, or to a Western diet and lifestyle. The removal of the appendix at an early age may be protective. Diagnosis is typically by colonoscopy with tissue biopsies. It is a kind of inflammatory bowel disease (IBD) along with Crohn's disease and microscopic colitis. Dietary changes, such as maintaining a high-calorie diet or lactose-free diet, may improve symptoms. Several medications are used to treat symptoms and bring about and maintain remission, including aminosalicylates as mesalazine or sulfasalazine, steroids, immunosuppressants such as azathioprine, and biologic therapy. Removal of the colon by surgery may be necessary if the disease is severe, does not respond to treatment, or if complications

such as colon cancer develop. Removal of the colon and rectum can cure the disease.

Causes of UC

No direct causes for UC are known, but many possible factors such as genetics and stress play a role.

Genetic factors

A genetic component to the cause of UC can be hypothesized based on:

- Aggregation of ulcerative colitis in families
- Identical twin concordance rate of 10% and dizygotic twin concordance rate of 3%
- Ethnic differences in the number of cases
- Genetic markers and linkages

Twelve regions of the genome may be linked to UC, including, in the order of their discovery, chromosomes 16, 12, 6, 14, 5, 19, 1, and 3, but none of these loci has been consistently shown to be at fault, suggesting that the disorder is influenced by multiple genes. For example, chromosome band 1p36 is one such region thought to be linked to inflammatory bowel disease.

Some of the putative regions encode transporter proteins such as OCTN1 and OCTN2. Other potential regions involve cell scaffolding proteins such as the MAGUK family. Human leukocyte antigen associations may even be at work. In fact, this linkage on chromosome 6 may be the most convincing and consistent of the genetic candidates.

Environmental factors

It is hypothesized that environmental factors contributing to the pathogenesis of ulcerative colitis. These are as below-

Diet: As the colon is exposed to many dietary substances which may be affected with inflammation. Dietary factors have been hypothesized to play a role in the pathogenesis of both ulcerative colitis

and Crohn's disease. Few studies have investigated such an association of dietary intake and genesis of IBS, which are as below.

1. One study showed no association of refined sugar on the number of people affected of ulcerative colitis.
2. High intake of unsaturated fat and vitamin B6 may enhance the risk of developing ulcerative colitis.
3. Other identified dietary factors that may influence the development and/or relapse of the disease include meat protein and alcoholic beverages.
4. Sulfur has been investigated as being involved in the cause of ulcerative colitis, but this is controversial. Sulfur restricted diets have been investigated in people with UC and animal models of the disease. The theory of sulfur as an etiological factor is related to the gut microbiota and mucosal sulfide detoxification in addition to the diet.

Breastfeeding: Some reports of the protection of breastfeeding in the development of IBD contradict each other. One Italian study showed a potential protective effect.

One study of isotretinoin found a small increase in the rate of UC.

UC as autoimmune disease

Ulcerative colitis is an autoimmune disease characterized by T-cells infiltrating the colon. In contrast to Crohn's disease, which can affect areas of the gastrointestinal tract outside of the colon, ulcerative colitis usually involves the rectum and is confined to the colon, with occasional involvement of the ileum. This so-called "backwash ileitis" can occur in 10-20% of people with pancolitis and is believed to be of little clinical significance. Ulcerative colitis can also be associated with comorbidities that produce symptoms in many areas of the body outside the digestive system. Surgical removal of the large intestine often cures the disease.

Alternative theories

Risk factors

	Crohn's disease	Ulcerative colitis
Smoking	Higher risk for smokers	Lower risk for smokers
Age	Usual onset between 15 and 30 years	Peak incidence between 15 and 25 years

Levels of sulfate-reducing bacteria tend to be higher in persons with ulcerative colitis, which could indicate higher levels of hydrogen sulfide in the intestine. An alternative theory suggests that the symptoms of the disease may be caused by toxic effects of the hydrogen sulfide on the cells lining the intestine.

Infection by mycobacterium avium, subspecies para-tuberculosis, has been proposed as the ultimate cause of both ulcerative colitis and Crohn's disease.

Types of Ulcerative Colitis

The type of ulcerative colitis you have depends on where it's located in your body:

- **Ulcerative proctitis:** It is usually the mildest form. It's only in the rectum, the part of colon closest to anus. Rectal bleeding may be the only sign of the disease.
- **Proctosigmoiditis:** Rectum and the lower end of colon are involved. In this condition patients have bloody diarrhea, belly cramps, and pain. Patients have the urge to poop but would not be able to.
- **Left-sided:** It causes cramps on that side of belly. Patients will also have bloody diarrhea, and might lose weight without trying. Patients will have inflammation from rectum up through the left side of colon.
- **Pancolitis:** It often affects entire colon. It can cause severe bouts of bloody diarrhea, belly cramps, pain, fatigue, and major weight loss.
- **Acute severe ulcerative colitis:** It is rare form of UC. It affects entire colon and causes severe pain, heavy diarrhea, bleeding and fever.

Pathophysiology

	Crohn's disease	Ulcerative colitis
Cytokine response	Associated with Th17	Vaguely associated with Th2

An increased amount of colonic sulfate-reducing bacteria has been observed in some people with ulcerative colitis, resulting in higher concentrations of the toxic gas hydrogen sulfide. Human colonic mucosa is maintained by the colonic epithelial barrier and immune cells in the lamina propria (see intestinal mucosal barrier). N-butyrate, a short-chain fatty acid, gets oxidized through the beta oxidation pathway into carbon dioxide and ketone bodies. It has been shown that N-butyrate helps supply nutrients to this epithelial barrier. Studies have proposed that hydrogen sulfide plays a role in impairing this beta-oxidation pathway by interrupting the short chain acetyl-CoA dehydrogenase, an enzyme within the pathway. Furthermore, it has been suggested that the protective benefit of smoking in ulcerative colitis is due to the hydrogen cyanide from cigarette smoke reacting with hydrogen sulfide to produce the non-toxic isothiocyanate, thereby inhibiting sulfides from interrupting the pathway. An unrelated study suggested that the sulfur contained in red meats and alcohol may lead to an increased risk of relapse for people in remission.

Patients with ulcerative colitis typically present with rectal bleeding, diarrhea, tenesmus (urgent desire to evacuate the bowels but with the passage of little stool), and lower abdominal pain. The severity of disease at clinical presentation is important in determining the appropriate therapy. Peoples with mildly active disease will have fewer than 4 bowel movements daily and no signs of toxicity. Individuals with moderate-severity UC have more frequent bowel movements with bleeding. Approximately 70% of people with ulcerative colitis will have moderately active disease at presentation. People with severely active disease will have signs of toxicity with fever, tachycardia, and anemia. People with fulminant or toxic colitis or toxic megacolon often have more than 10 bowel movements in a day, continuous bleeding,

abdominal distention and tenderness, and radiologic evidence of edema and, in some cases, bowel dilation. These people most often require immediate colectomy because 10% have perforated colon at the time of surgery,

Signs and symptoms

Items	Crohn's disease	Ulcerative colitis
Defecation	Often porridge-like, sometimes steatorrhea	Often mucus-like and with blood
Tenesmus	Less common	More common
Fever	Common	Indicates severe disease
Fistulae	Common	Seldom
Weight loss	Often	More seldom

Gastrointestinal

The clinical presentation of ulcerative colitis depends on the extent of the disease process. Patients with ulcerative colitis usually present with diarrhea mixed with blood and mucus, of gradual onset that persists for an extended period (weeks). They may also have weight loss and blood on rectal examination. The inflammation caused by the disease along with the chronic bleeding from the GI tract leads to increased rates of anemia. The disease may be accompanied by different degrees of abdominal pain, from mild discomfort to painful bowel movements or painful abdominal cramping with bowel movements.

Ulcerative colitis is associated with a general inflammatory process that can affect many parts of the body. Sometimes, these associated extra-intestinal symptoms are the initial signs of the disease, such as painful arthritic knees in teenagers, which also may be seen in adults. A diagnosis of UC may not occur until the onset of intestinal manifestations.

Extent of intestinal involvement

Ulcerative colitis is normally continuous from the rectum up the colon. The disease is classified by the extent of involvement, depending on how far the disease extends:

- Distal colitis, potentially treatable with enemas:

- o Proctitis: Involvement limited to the rectum
- o Proctosigmoiditis: Involvement of the rectosigmoid colon, the portion of the colon adjacent to the rectum
- o Left-sided colitis: Involvement of the descending colon, which runs along the person's left side, up to the splenic flexure and the beginning of the transverse colon
- Extensive colitis: Inflammation extending beyond the reach of enemas:
- o Pancolitis: Involvement of the entire colon, extending from the rectum to the cecum, beyond which the small intestine begins

In addition to the extent of involvement, people may also be characterized by the severity of their disease.

- Mild disease correlates with fewer than four stools daily, with or without blood, no systemic signs of toxicity, and a normal erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP). Mild abdominal pain or cramping may occur. People with UC may believe they are constipated when in fact they are experiencing tenesmus, which is a constant feeling of the need to empty the bowel accompanied by involuntary straining efforts, pain, and cramping with little or no fecal output.
- Moderate disease correlates with more than four stools daily, but with minimal signs of toxicity. People may display anemia (not requiring transfusions), moderate abdominal pain, and low-grade fever, 38 to 39 °C (100 to 102 °F).
- Severe disease correlates with more than six bloody stools a day or observable massive and significant bloody bowel movement, and evidence of toxicity as demonstrated by fever, tachycardia, anemia or an elevated ESR or CRP.

Fulminant disease correlates with more than 10 bowel movements daily, continuous bleeding, toxicity, abdominal tenderness and distension, blood transfusion requirement, and colonic dilation (expansion). People in this category may have inflammation extending beyond just the mucosal

layer, causing impaired colonic motility and leading to toxic megacolon. If the serous membrane is involved, a colonic perforation may ensue. Unless treated, the fulminant disease will soon lead to death.

Ulcerative colitis - Disease severity

Classification	Description	Prevalence
Mild	• <4 stools/day +/- blood • Normal ESR • No signs for systemic toxicity	22%-28%
Moderate	• ≥4 stools/day • Minimal signs of toxicity	43%-46%
Severe	• >6 blood stools/day • Fever, tachycardia, anemia, or / ESR	28%-31%
Fulminant	• >10 stools/day, continuous bleeding • Toxicity, abdominal tenderness/distension, transfusion requirement, colonic dilation on x-ray	~9%

ESR = erythrocyte sedimentation rate.

Extra-intestinal manifestation

As UC is believed to have a systemic (i.e., autoimmune) origin, people with UC may present with comorbidities leading to symptoms and complications outside the colon. The frequency of such extra-intestinal manifestations has been reported as between 6 and 47% and it includes:

- Aphthous ulcer of the mouth
- Ophthalmic
 - o Iritis or uveitis, which is inflammation of the eye's iris
 - o Episcleritis
- Musculoskeletal:
 - o Seronegative arthritis, which can be a large-joint oligoarthritis (affecting one or two joints), or may affect many small joints of the hands and feet
 - o Ankylosing spondylitis, arthritis of the spine
 - o Sacroiliitis, arthritis of the lower spine
- Cutaneous (related to the skin):

- o Erythema nodosum, which is a panniculitis, or inflammation of subcutaneous tissue involving the lower extremities
- o Pyoderma gangrenosum, which is a painful ulcerating lesion involving the skin
- Deep venous thrombosis and pulmonary embolism
- Autoimmune hemolytic anemia
- Clubbing, a deformity of the ends of the fingers
- Primary sclerosing cholangitis, a distinct disease that causes inflammation of the bile ducts

Irritable bowel syndrome

Irritable bowel syndrome (IBS) is a syndromic presentation of group of symptoms-including abdominal pain and changes in the pattern of bowel movements without any evidence of underlying cause. These symptoms occur over a long time, often years. It has been classified into four main types depending on whether diarrhea is common, constipation is common, both are common, or neither occurs very often (IBS-D, IBS-C, IBS-M, or IBS-U respectively). IBS negatively affects quality of life and may result in missed school or work. Disorders such as anxiety, major depression, and chronic fatigue syndrome are common among people with IBS.

In conventional system of medicine the causes of IBS are not clear. Theories include combinations of gut-brain axis problems, gut motility disorders, pain sensitivity, infections including small intestinal bacterial overgrowth, neurotransmitters, genetic factors, and food sensitivity. Onset may be triggered by an intestinal infection, or stressful life event. IBS is a functional gastrointestinal disorder. Diagnosis is based on symptoms in the absence of worrisome features and once other potential conditions have been ruled out. Worrisome features include onset at greater than 50 years of age, weight loss, blood in the stool, or a family history of inflammatory bowel disease. Other conditions that may present similarly include celiac disease, microscopic colitis, inflammatory bowel disease, bile acid malabsorption, and colon cancer.

There is no known cure for IBS in modern medicine. Treatment is carried out to improve symptoms. This may include dietary changes, medication, probiotics, and counseling. Dietary measures include increasing soluble fiber intake, a gluten-free diet, or a short-term diet low in fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAPs). The medication loperamide may be used to help with diarrhea while laxatives may be used to help with constipation. Antidepressants may improve overall symptoms and reduce pain. Patient education and a good doctor-patient relationship are an important part of care.

About 10 to 15% of people in the developed world are believed to be affected by IBS. It is estimated that 45% of people globally are affected by IBS. It is more common in South America and less common in Southeast Asia. It is twice as common in women as men and typically occurs before age 45. The condition appears to become less common with age. IBS does not affect life expectancy or lead to other serious diseases. The first description of the condition was in 1820 while the current term "irritable bowel syndrome" came into use in 1944.

Women are around two to three times more likely to be diagnosed with IBS and four to five times more likely to seek specialty care for it than men. These differences likely reflect a combination of both biological (sex) and social factors. Finally, sexual trauma is a major risk factor for IBS, with as many as 33% of those affected reporting such abuse. Because women are at higher risk of sexual abuse than men, sex-related risk of abuse may contribute to the higher rate of IBS in women.

Cause of IBS

The risk of developing IBS increases six-fold after acute gastrointestinal infection. Post-infection, further risk factors are young age, prolonged fever, anxiety, and depression. Psychological factors, such as depression or anxiety, have not been shown to cause or influence the onset of IBS, but may play significant role in the persistence and perceived severity of symptoms.

1. **Stress :** It is believed that psychological stress may trigger IBS in predisposed individuals. The stress response in the body involves the hypothalamic-pituitary-adrenal axis (HPA) and the sympathetic nervous system, both of which have been shown to operate abnormally in people with IBS. Psychiatric illness or anxiety precedes IBS symptoms in two-thirds of people with IBS, and psychological traits predispose previously healthy people to developing IBS after gastroenteritis.
2. **Post-infectious:** Approximately 10 percent of IBS cases are triggered by an acute gastroenteritis infection. Genetic defects relating to the innate immune system and epithelial barrier as well as high stress and anxiety levels appear to increase the risk of developing post-infectious IBS. Post-infectious IBS usually manifests itself as the diarrhea-predominant subtype. However, increased gut permeability is strongly associated with IBS regardless of whether IBS was initiated by an infection or not. A link between small intestinal bacterial overgrowth and tropical sprue has been proposed to be involved as a cause of post-infectious IBS.
3. **Bacteria:** Small intestinal bacterial overgrowth (SIBO) occurs with greater frequency in people who have been diagnosed with IBS compared to healthy controls. Symptoms of SIBO include bloating, abdominal pain, diarrhea or constipation among others. IBS may be the result of the immune system interacting abnormally with gut microbiota resulting in an abnormal cytokine signaling profile.
4. **Fungus:** The role of the gut mycobiota and especially of the abnormal proliferation of the yeast *Candida albicans* in some people with IBS is found.
5. **Protozoa:** Prevalence of protozoal infections is greater in industrialized countries (United States and Canada) in the 21st century. Evidences suggest that blastocystis-colonization occurs more commonly in IBS affected individuals and is a possible risk factor

for developing IBS. *Dientamoeba fragilis* has also been considered a possible organism to study, though it is also found in people without IBS.

6. **Vitamin D:** Vitamin D deficiency is more common in individuals affected by irritable bowel syndrome. Vitamin D is involved in regulating triggers for IBS including the gut microbiome, inflammatory processes and immune responses, as well as psychosocial factors.
7. **Genetics:** SCN5A mutations are found in a small number of people who have IBS, particularly the constipation-predominant variant (IBS-C). The resulting defect leads to disruption in bowel function, by affecting the Nav1.5 channel, in smooth muscle of the colon and pacemaker cells.

Mechanism

Genetic, environmental, and psychological factors seem to be important in the development of IBS. Studies have shown that IBS has a genetic component even though there is a predominant influence of environmental factors.

There is evidence that abnormalities occur in the gut flora of individuals who have IBS, such as reduced diversity, a decrease in bacteria belonging to the phylum *Bacteroidetes*, and an increase in those belonging to the phylum *Firmicutes*. The changes in gut flora are most profound in individuals who have diarrhoea predominant IBS. Antibodies against common components (namely flagellin) of the commensal gut flora are a common occurrence in IBS affected individuals.

Chronic low-grade inflammation commonly occurs in IBS affected individuals with abnormalities found including increased enterochromaffin cells, intraepithelial lymphocytes, and mast cells resulting in chronic immune-mediated inflammation of the gut mucosa. IBS has been reported in greater quantities in multigenerational families with IBS than in the regular population. It is believed that psychological stress can induce increased inflammation and thereby cause IBS to develop in predisposed individuals.

Signs and symptoms

- Abdominal pain or discomfort in association with frequent diarrhea or constipation and a change in bowel habits, these are experienced as acute attacks that subside within one day, but recurrent attacks are common.
- Urgency for bowel movements,
- A feeling of tenesmus or bloating.
- Some cases, the symptoms are relieved by bowel movements.
- People with IBS, more commonly than others, have gastroesophageal reflux, symptoms relating to the genitourinary system, chronic fatigue, syndrome, fibromyalgia, headache, backache, and psychiatric symptoms such as depression and anxiety.
- About a third of men and women who have IBS also report sexual dysfunction typically in the form of a reduction in libido.

Food intolerance

Symptoms in up to 90% of people are brought on by food intolerances. These often include fermentable oligosaccharides, disaccharides, monosaccharides and polyols (FODMAPs)

Diagnosis

No specific laboratory or imaging tests can diagnose irritable bowel syndrome. Diagnosis should be based on symptoms. Rome criteria are usually used to diagnose the cases of IBS. They allow the diagnosis to be based only on symptoms, but no criteria based solely on symptoms is sufficiently accurate to diagnose IBS. Worrisome features include onset at greater than 50 years of age, weight loss, blood in the stool, iron-deficiency anemia, or a family history of colon cancer, celiac disease, or inflammatory bowel disease.

Rome criteria

The Rome IV criteria for the diagnosis of irritable bowel syndrome require that patients have had recurrent abdominal pain on average at least 1 day per week during the previous 3 months that is associated with two or more of the following:

- Related to defecation (may be increased or unchanged by defecation)

- Associated with a change in stool frequency
- Associated with a change in stool form or appearance

The Rome IV criteria (May 2016) only require abdominal pain in defining this condition; "discomfort" is no longer a requirement owing to its non-specificity, and the recurrent abdominal pain.

Supporting symptoms include the following:

- Altered stool frequency
- Altered stool form
- Altered stool passage (straining and/or urgency)
- Mucorrhea
- Abdominal bloating or subjective distention

Physicians may choose to use one of these guidelines or may simply choose to rely on their own anecdotal experience with past patients. The algorithm may include additional tests to guard against misdiagnosis of other diseases as IBS. Such "red flag" symptoms may include weight loss, gastrointestinal bleeding, anemia, or nocturnal symptoms. However, red flag conditions may not always contribute to accuracy in diagnosis; for instance, as many as 31% of people with IBS have blood in their stool, many possibly from hemorrhoidal bleeding.

IBS Subtypes: IBS is classified into four subtypes such as-

1. IBS with predominant constipation (IBS-C)
2. IBS with predominant diarrhea (IBS-D)
3. IBS with mixed bowel habits (IBS-M)
4. IBS of untyped.

One important change from Rome III, as noted above, is that the IBS subtype is explicitly based on the patient's reported predominant bowel habit on days with abnormal bowel movements, and not on an average of all days, which might include days with normal bowel habits. Abnormal bowel movements are classified using the Bristol stool form scale, which is described below. For clinical trials, or when appropriate in clinical settings, subjects should complete a 14-day bowel diary to

most accurately categorize IBS subtypes. Bristol stool types 1 and 2 or types 6 and 7 are considered abnormal.

Bristol Stool Form Scale

The Bristol stool form scale (BSFS) was developed in the 1990s in the Bristol Royal Infirmary in England. The authors described seven types of stool, which are noted below:

- Type 1: Separate hard lumps, like nuts (hard to pass)
- Type 2: Sausage-shaped, but lumpy
- Type 3: Like a sausage but with cracks on its surface
- Type 4: Like a sausage or snake, smooth and soft
- Type 5: Soft blobs with clear cut edges (passed easily)
- Type 6: Fluffy pieces with ragged edges, a mushy stool
- Type 7: Watery, no solid pieces, entirely liquid

Stool types 1 and 2 as being associated with constipation, while stool types 6 and 7 were associated with diarrhea (and stool type 5 to some degree). Stool types 3 and 4 were considered normal stools. The BSFS is a convenient way for patients to describe their bowel habits, and is routinely used in clinical trials.

Differential diagnosis

Differential Diagnosis of IB

IBS with diarrhea

Dietary - lactose, sorbitol, fructose, caffeine, alcohol, fatty foods, fat substitutes, gas-producing foods

Infections- *Giardia* spp, *Amoeba* spp, HIV-related, bacterial overgrowth

IBD- Crohn's disease, ulcerative colitis, microscopic colitis

Drug toxicity- antibiotics, proton pump inhibitors, nonsteroidal anti-inflammatory drugs, ACE inhibitors, beta-blockers, chemotherapy

Malabsorption- celiac disease, bile acid-related

Other- ovarian cancer, endometriosis, colorectal cancer, hyperthyroidism, carcinoid, VIPoma, ischemic colitis

IBS with constipation

Dietary/mode of life- inadequate fiber, immobility

Neurologic- Parkinson's disease, multiple sclerosis, spinal cord injuries

Endocrine- diabetes, hypothyroidism, hypercalcemia

Drug toxicity- opiate analgesics, calcium-channel blockers, antidepressants, clonidine

Other - colorectal cancer, ovarian cancer, bowel obstruction, diverticular disease, endometriosis

Investigations

Investigations are performed to exclude other clinical conditions:

- Stool microscopy and culture (to exclude infectious conditions)
- Blood tests: Full blood examination, liver function tests, erythrocyte sedimentation rate, and serological testing for coeliac disease
- Abdominal ultrasound (to exclude gallstones and other biliary tract diseases)
- Endoscopy and biopsies (to exclude peptic ulcer disease, coeliac disease, inflammatory bowel disease, and malignancies)
- Hydrogen breath testing (to exclude fructose and lactose malabsorption)

Misdiagnosis

IBS patients are at increased risk of being given inappropriate surgeries such as appendectomy, cholecystectomy, and hysterectomy due to being misdiagnosed as other medical conditions. Some common examples of misdiagnosis include infectious diseases, coeliac disease, *Helicobacter pylori*, parasites (non-protozoal). The American College of Gastroenterology recommends all people with symptoms of IBS be tested for coeliac disease.

Bile acid malabsorption is also sometimes

missed in people with diarrhea-predominant IBS. SeHCAT tests suggest around 30% of people with D-IBS have this condition, and most respond to bile acid sequestrants.

Chronic use of certain sedative-hypnotic drugs, especially the benzodiazepines, may cause irritable bowel-like symptoms that can lead to a misdiagnosis of irritable bowel syndrome.

Comorbidities

Several medical conditions, or comorbidities, appear with greater frequency in people with IBS.

- **Neurological/psychiatric:** IBS occurs in 51% of people with chronic fatigue syndrome and 49% of people with fibromyalgia, and psychiatric disorders occur in 94% of people with IBS.
- **Inflammatory bowel disease:** IBS may be a type of low-grade inflammatory bowel disease. Researchers have suggested IBS and IBD are interrelated diseases, noting that people with IBD experience IBS-like symptoms when their IBD is in remission.
- **Abdominal surgery:** People with IBS were at increased risk of having unnecessary gall bladder removal, abdominal, pelvic surgery etc.
- **Endometriosis:** There is a link between migraine headaches, IBS, and endometriosis.

- **Other chronic disorders:** Interstitial cystitis may be associated with other chronic pain syndromes, such as irritable bowel syndrome and fibromyalgia.

Inflammatory Bowel Disease

1. Ulcerative colitis and Crohn's disease are chronic inflammatory bowel diseases which pursue a protracted relapsing and remitting course, usually extending over years.
2. The diseases have many similarities and it is sometimes impossible to differentiate between them.
3. Ulcerative colitis only involves the colon
4. Crohn's disease can involve any part of the gastrointestinal tract from mouth to anus.
5. Crohn's disease appears to be very rare in the developing world yet ulcerative colitis, although still unusual, is becoming more common.
6. Life expectancy in patients with IBD is similar to that of the general population
7. Although many patients require surgery and admission to hospital for other reasons, the majority have an excellent work record and pursue a normal life.
8. Around 90% of ulcerative colitis patients have intermittent disease activity, whilst 10%, have continuous symptoms.

Comparison of ulcerative colitis and Crohn's disease

	Ulcerative colitis	Crohn's disease
Age group	Any	Any
Gender	M = F	M = F
Ethnic group	Any <i>HLA-DR103</i> associated with	Any; more common in Ashkenazi Jews
Genetic factors	severe disease	<i>CARD 15/NOD-2</i> mutations predispose
Risk factors	More common in non-/ex-smokers Appendicectomy protects	More common in smokers
Anatomical distribution	Colon only; begins at anorectal margin with variable proximal extension	Any Part of gastrointestinal tract; perianal disease common; patchy distribution- 'skip lesions'
Extraintestinal manifestations	Common	Common

Presentation	Bloody diarrhoea	Variable; pain, diarrhoea, weight loss all common
Histology	Inflammation limited to mucosa; crypt distortion, cryptitis, crypt abscesses, loss of goblet cells	Submucosal or transmural inflammation common; deep fissuring ulcers, fistulas; patchy changes; granulomas
Management	5-ASA; corticosteroids;	Corticosteroids; methotrexate; biologic
	azathioprine; colectomy is curative	therapy (anti-TNF); nutritional therapy; surgery for complications is not curative

Pathophysiology

It is thought that IBD develops because of an abnormal host response to an environmental trigger in genetically susceptible individuals



This causes inflammation of the intestine and release of inflammatory mediators, such as TNF, IL-12 and IL-23



Which cause tissue damage.



These mediators are targets for therapeutic intervention



In both diseases the intestinal wall is infiltrated with acute and chronic inflammatory cells



Inflammatory Bowel Disease

Factors associated with the development of inflammatory bowel disease

Genetic

- More common in Ashkenazi Jews
- 10% have a first-degree relative or at least one close relative with IBD
- High concordance between identical twins
- Both ulcerative colitis and Crohn's disease are associated with the HLA locus and the genes encoding MST-1 and IL-23
- Crohn's disease is associated with the *NOD2*, *ATG16L1* and *IRGM* genes
- Ulcerative colitis is associated with the *ECM-1* gene

- Associated with autoimmune thyroiditis and SLE
- HLA-DR103 associated with severe ulcerative colitis
- Ulcerative colitis and Crohn's patients with HLA-B27 commonly develop ankylosing spondylitis

Environmental

- Ulcerative colitis is more common in non-smokers and ex-smokers
- Most Crohn's patients are smokers (relative risk = 3)
- Associated with low-residue, high refined sugar diet
- Appendectomy protects against ulcerative colitis

Clinical features-Ulcerative colitis

1. The major symptom is bloody diarrhoea.
2. The first attack is usually the most severe and thereafter the disease is followed by relapses and remissions.
3. Emotional stress, intercurrent infection, gastroenteritis, antibiotics or NSAID therapy may all provoke a relapse.
4. Proctitis causes rectal bleeding and mucus discharge, sometimes accompanied by tenesmus.
5. Some patients pass frequent, small-volume fluid stools, while others are constipated and pass pellety stools.
6. Constitutional symptoms do not occur.

7. Proctosigmoiditis causes bloody diarrhoea with mucus.
8. Almost all patients are constitutionally well, but a small minority who have very active, limited disease develop fever, lethargy and abdominal discomfort.
9. Extensive colitis causes bloody diarrhoea with passage of mucus.
10. In severe cases anorexia, malaise, weight loss and abdominal pain occur, and the patient is toxic with fever, tachycardia and signs of peritoneal inflammation

Factors considered for disease severity assessment in ulcerative colitis

	Mild	Severe
Daily bowel frequency	< 4	> 6
Blood in stools	+/-	+++
Stool volume	< 200 g/24 hrs	> 400 g/24 hr.
Pulse	< 90 bpm	> 90 bpm
Temperature	Normal	> 37.8°C, 2 days out of 4
Sigmoidoscopy	Normal or granular mucosa	Blood in lumen
Abdominal X-ray	Normal	Dilated bowel and/or mucosal islands
Haemoglobin	Normal	< 100 g/L (< 10 g/dL)
ESR	Normal	> 30 mm/hr
Serum albumin	> 35 g/L (> 3.5 g/dL)	< 30 g/L (< 3 g/dL)

Crohn's disease

1. The major symptoms are abdominal pain, diarrhoea and weight loss.
2. Real Crohn's disease may cause subacute or even acute intestinal obstruction.
3. The pain is often associated with diarrhoea which is usually watery and does not contain blood or mucus.
4. Almost all patients lose weight because they avoid food since eating provokes pain.
5. Weight loss may also be due to malabsorption,

and some patients present with features of fat, protein or vitamin deficiencies.

6. Crohn's colitis presents in an identical manner to ulcerative colitis, but rectal sparing and the presence of perianal disease are features which favour a diagnosis of Crohn's disease.
7. Many patients present with symptoms of both small bowel and colonic disease.
8. A few have isolated perianal disease, vomiting from jejunal strictures or severe oral ulceration.

Physical examination

1. Physical examination often reveals
 - Evidence of weight loss
 - Anaemia
 - Glossitis
 - Angular stomatitis.
 - Abdominal tenderness most marked over the inflamed area.
 - An abdominal mass due to matted loops of thickened bowel or an intra- abdominal abscess may occur.
 - Perianal skin tags, fissures or fistulas are found in at least 50% of patients.

Investigations

Blood :

1. Full blood count
 - Anaemia resulting from bleeding or malabsorption of iron, folic acid or vitamin B12.
2. Serum albumin concentration falls as a consequence of protein-losing enteropathy, inflammatory disease or poor nutrition.
3. The ESR and CRP are elevated in exacerbations and in response to abscess formation.

Bacteriology :

4. Stool microscopy, culture and examination for *Clostridium difficile* toxin or for ova and cysts,
5. Blood cultures and serological tests should be performed to exclude infection.

Endoscopy :

6. Sigmoidoscopy with biopsy is an essential

investigation in all patients who present with diarrhoea.

7. Colonoscopy may show active inflammation with pseudopolyps or a complicating carcinoma.
8. Biopsies are confluent and most severe in the distal colon and rectum.
9. Capsule endoscopy is useful in the identification of small bowel inflammation, but should be avoided in the presence of strictures.

Radiology :

10. A plain abdominal X-ray is essential in the management of patients who present with severe active disease.
 - Dilatation of the colon, mucosal oedema ('thumb-printing') or evidence of perforation may be found.
 - In small bowel Crohn's disease there may be evidence of intestinal obstruction or displacement of bowel loops by a mass.
11. Barium enema is a less sensitive investigation than colonoscopy in patients with colitis and is now little used.
 - Contrast studies of the small bowel can be helpful in the investigation of Crohn's disease; affected areas are narrowed and ulcerated, and multiple strictures are common

MRI scans :

12. MRI scans provide comparable information without radiation exposure or risking small bowel obstruction and are useful in delineating pelvic or perineal involvement.

Ultrasound :

13. Ultrasound may identify thickened small bowel loops and abscess development in Crohn's disease.

MCQs

1. Duodenal Ulcers occur most often in the
 - a. First portion of the duodenum (95%), with 90% located within 3 cm of the pylorus and they are usually ≤ 1 cm in diameter but can occasionally reach 3–6 cm (giant ulcer).
 - b. Second portion of the duodenum (95%),

- with 90% located within 4 cm of the pylorus and they are usually ≤ 2 cm in diameter but can occasionally reach 4–8 cm (giant ulcer).
- c. First portion of the duodenum (95%), with 90% located within 2 cm reach 2–5 cm (giant ulcer).
- d. Second portion of the duodenum (95%), with 90% located within 1 cm of the pylorus and they are usually ≤ 3 cm in diameter but can occasionally reach 6–8 cm (giant ulcer).
2. The Rome criteria specify that symptoms that characterize IBS must have been present for the previous 3 months. Which of the following is one of the Rome criteria for diagnosing irritable bowel syndrome (IBS)?
 - a. Flatulence
 - b. Nausea
 - c. Abdominal pain 3 or more days per week
 - d. Pain with defecation
3. During physical examination of a patient with suspected IBS, which abdominal location is most likely to be tender to palpation?
 - a. Left upper quadrant
 - b. Right upper quadrant
 - c. Left lower quadrant
 - d. Right lower quadrant
4. Which of the following infections has been associated with an increased prevalence of IBS?

a. Giardia lamblia	b. Escherichia coli
c. Shigella	d. Salmonella
5. Which of the following is a symptom consistent with a diagnosis of IBS?
 - a. Painless diarrhea
 - b. Fever
 - c. Postprandial urgency
 - d. Steatorrhea
6. Which of the following is not recognized as a symptom that supports the diagnosis of IBS according to the Rome criteria?
 - a. Altered stool frequency
 - b. Mucorrhea
 - c. Abdominal bloating or subjective distention
 - d. Frequent nausea

7. Which of the following symptoms indicates a need for laboratory testing or diagnostic imaging in patients with IBS younger than 50 years?
- a. Iron deficiency anemia
 - b. Abdominal pain
 - c. Amenorrhea
 - d. Hypokalemia
8. Which of the following is accurate about endoscopic findings in PUD and related conditions?

- a. Most gastric ulcers tend to occur at the junction of the fundus and antrum, along the lesser curvature
- b. Most duodenal ulcers are found in the second and third parts of the duodenum
- c. Malignant ulcers tend to have a smooth, regular, rounded edge with a flat, smooth base and surrounding mucosa that shows radiating folds
- d. The most common finding in individuals undergoing endoscopy for dyspepsia is peptic ulcer

Answer

1.	a	2.	d	3.	c
4.	a	5.	c	6.	d
7.	a	8.	a		